

Measuring & Decomposing Downside Risk in U.S. Healthcare Equities

A reproducible study of healthcare portfolio construction, factor exposure, and crisis behavior

Shuying Dai · 28 large-cap healthcare names · 2016–2025 (119 months) · Benchmarks: XLV, IBB, SPY

Bottom line. Within a sector treated as monolithically "defensive," risk is highly heterogeneous: annualized volatility ranges roughly 4× (JNJ 18% → MRNA 72%), tails are fat, and about half of portfolio risk is idiosyncratic. Transparent construction materially reshapes the risk profile — and the benefit is concentrated in rate-driven and biotech-specific stress, not broad market crashes.

Executive summary

This study builds three transparent healthcare portfolios — equal-weight (EW), minimum-variance (MV), and risk-parity (RP) — with a walk-forward monthly rebalance, compares them to sector and market benchmarks, decomposes their risk with a Fama-French 5 + momentum model, and stress-tests them through historical and hypothetical shocks. The emphasis is risk *explanation* and attribution, not alpha.

- **Risk-parity is the efficient construction** (Sharpe 0.91); **minimum-variance has the lowest volatility** (16.2%) but the deepest drawdown (−29.3%) — volatility ≠ tail risk.
- **~50% of portfolio variance is idiosyncratic**; the factor model explains only 44–50%, with no statistically significant alpha.
- **Defensive construction paid off in the 2022 rate selloff**: minimum-variance −3.9% vs the S&P 500 −24.1%.

Portfolio performance (2016–2025)

Portfolio	Ann. return	Ann. vol	Sharpe	Max DD	CVaR ₉₅
Equal-weight	14.7%	17.4%	0.88	−25.6%	2.5%
Minimum-variance	12.4%	16.2%	0.81	−29.3%	2.3%
Risk-parity	14.9%	16.9%	0.91	−25.7%	2.5%
XLV (sector)	10.8%	16.6%	0.70	−28.4%	2.4%
IBB (biotech)	7.0%	23.6%	0.40	−39.8%	3.3%
SPY (market)	15.6%	18.0%	0.90	−33.7%	2.8%

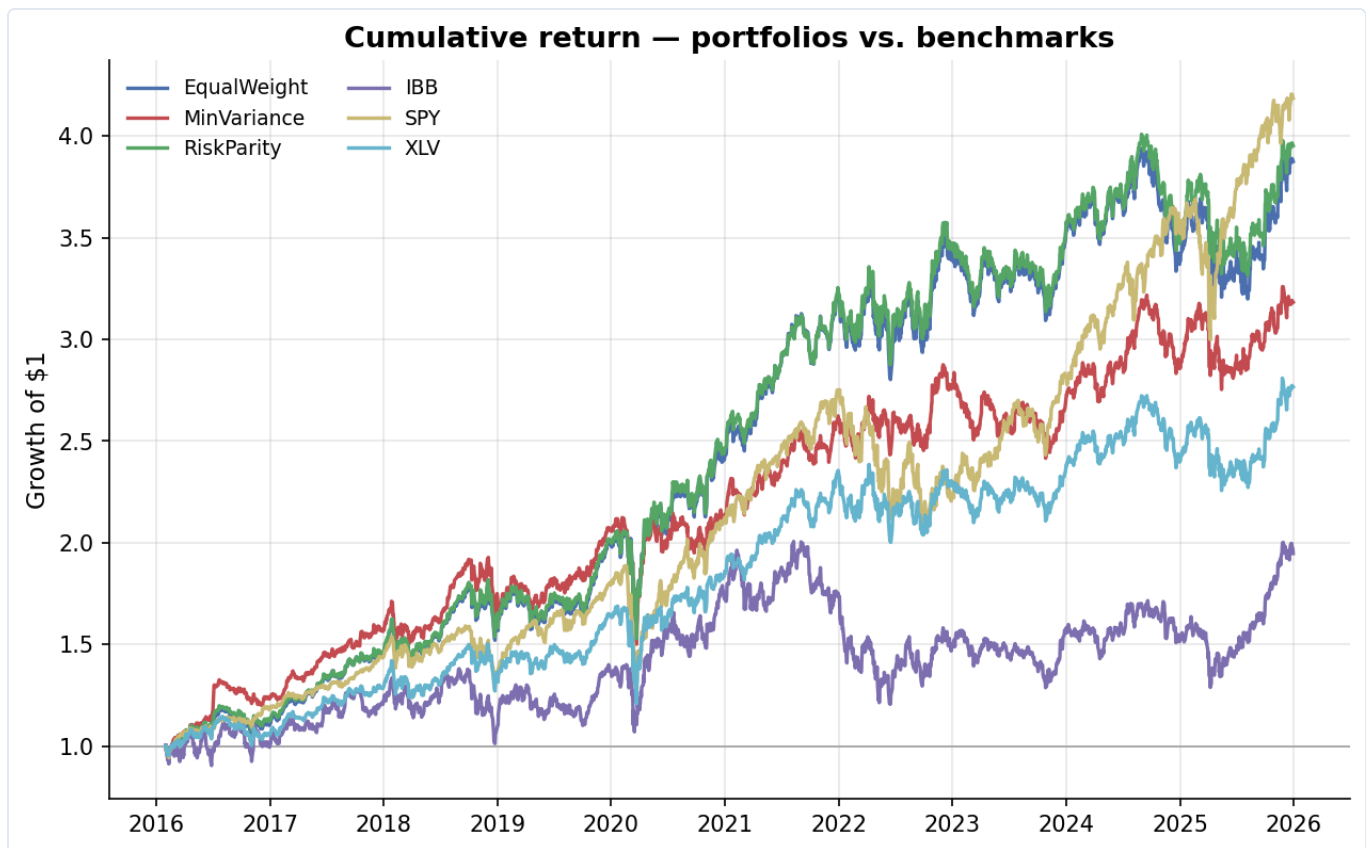


Figure 1. Growth of \$1 — the three healthcare portfolios vs. XLV, IBB and SPY. Risk-parity and equal-weight compound competitively with the market at lower volatility; biotech (IBB) lags with the deepest drawdowns.

Methodology

A reproducible, snapshot-based pipeline: daily adjusted prices (Yahoo Finance, stooq fallback), aligned to a master NYSE calendar with an explicit no-silent-fills missing-data policy. Portfolio weights are estimated **walk-forward** from a trailing 252-day covariance and rebalanced monthly (no look-ahead). Factor exposures use monthly Fama-French 5 + momentum (Kenneth French) with Newey-West (HAC) standard errors. Tail risk is reported as historical VaR and CVaR (Expected Shortfall) rather than Gaussian VaR, because every series is fat-tailed. The full stack is covered by 70 unit tests.

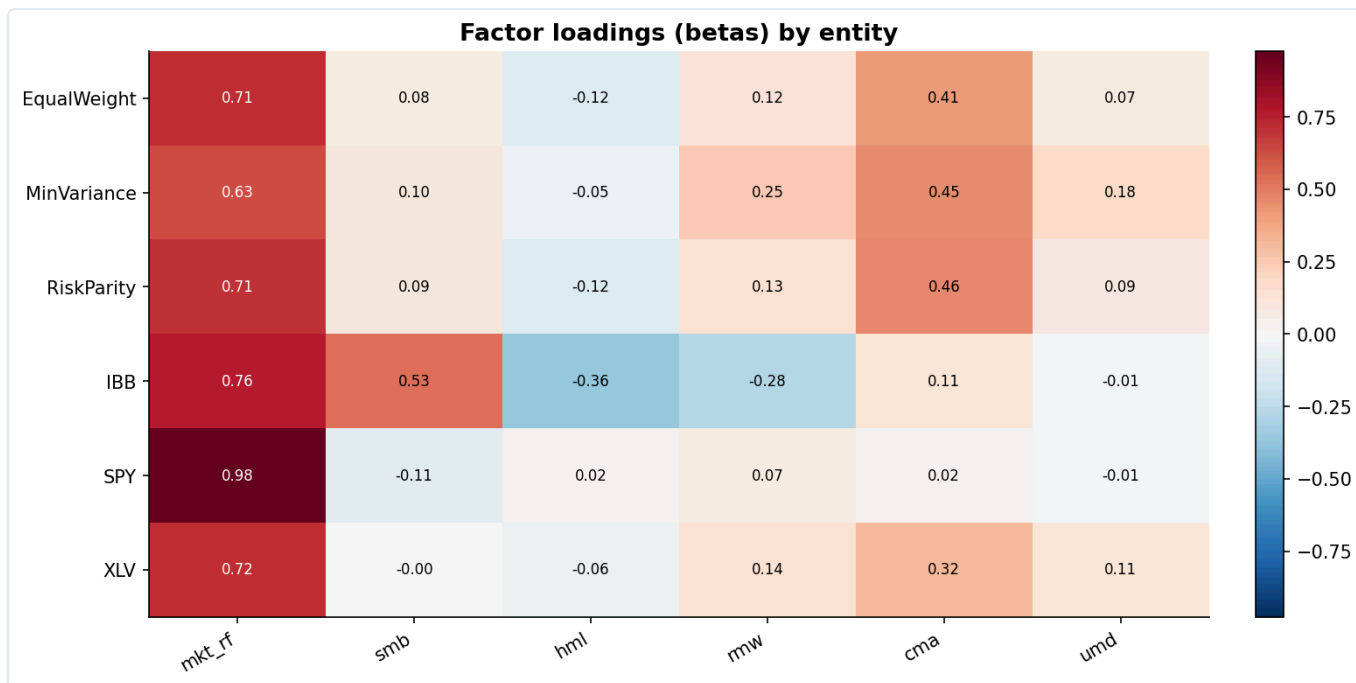


Figure 2. Factor loadings (betas) by entity. The model validity check passes — SPY loads ~0.98 on the market with $R^2 \approx 1.00$. Biotech (IBB) shows its signature small-cap (SMB +0.53) and growth (HML -0.36) tilt; minimum-variance carries the lowest market beta and a quality (RMW) tilt.

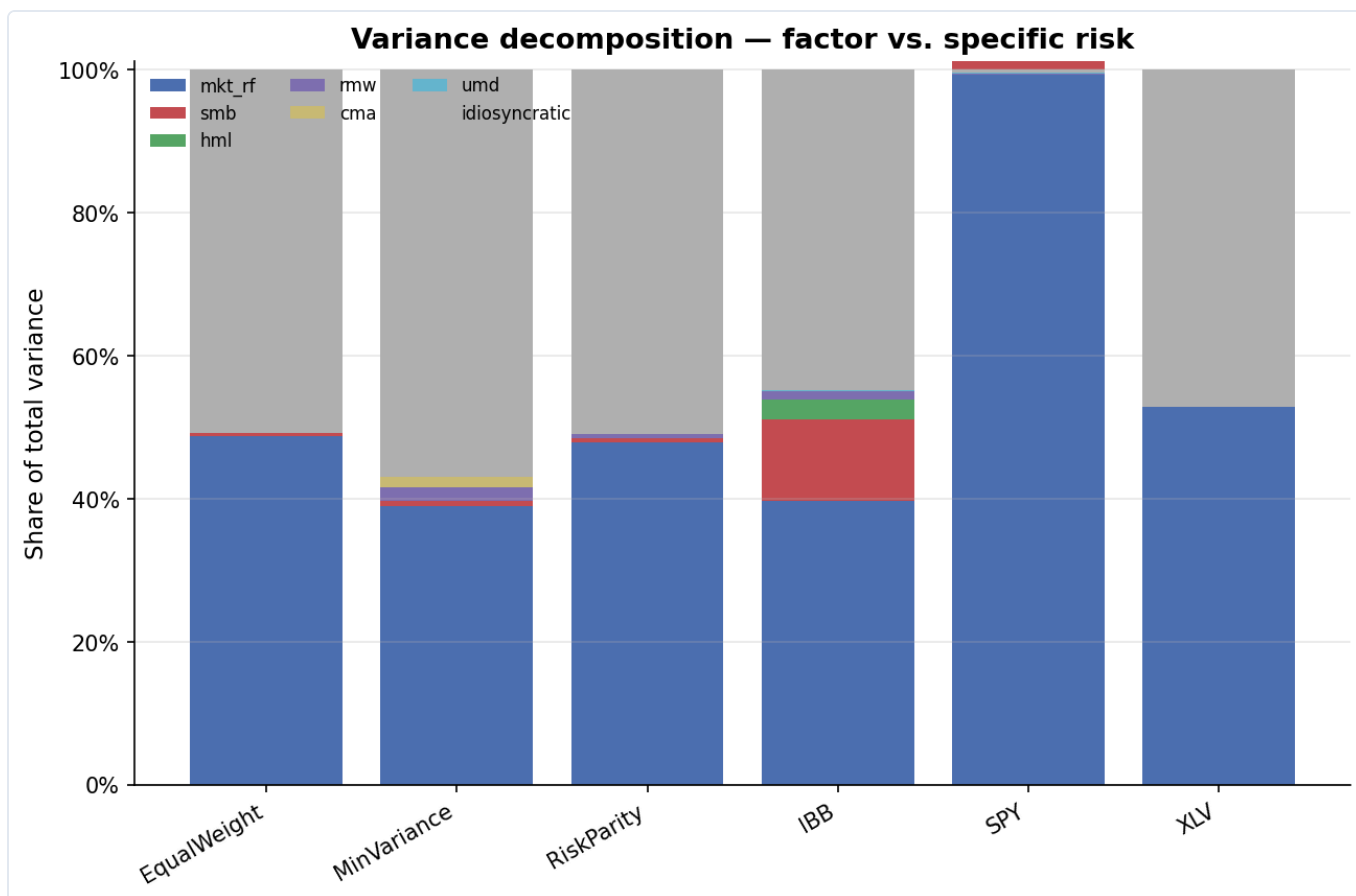


Figure 3. Variance decomposition — systematic (per factor) vs. idiosyncratic. Roughly half of each portfolio's variance is name-specific, consistent with healthcare's exposure to single-name catalysts (trials, approvals, pricing).

Stress testing

Two transparent tools: historical replay over labeled crisis windows, and hypothetical shocks propagated through linear betas (impact = $\beta \cdot \text{shock}$). In the broad COVID crash, healthcare and the market converged (−25% to −33%). In the 2022 rate-driven selloff the defensive portfolios decoupled hard — minimum-variance −3.9% vs SPY −24.1% — and in the 2015–16 biotech / drug-pricing selloff, diversified healthcare (−3.5% to −5.6%) avoided the worst of IBB's −8.6%.

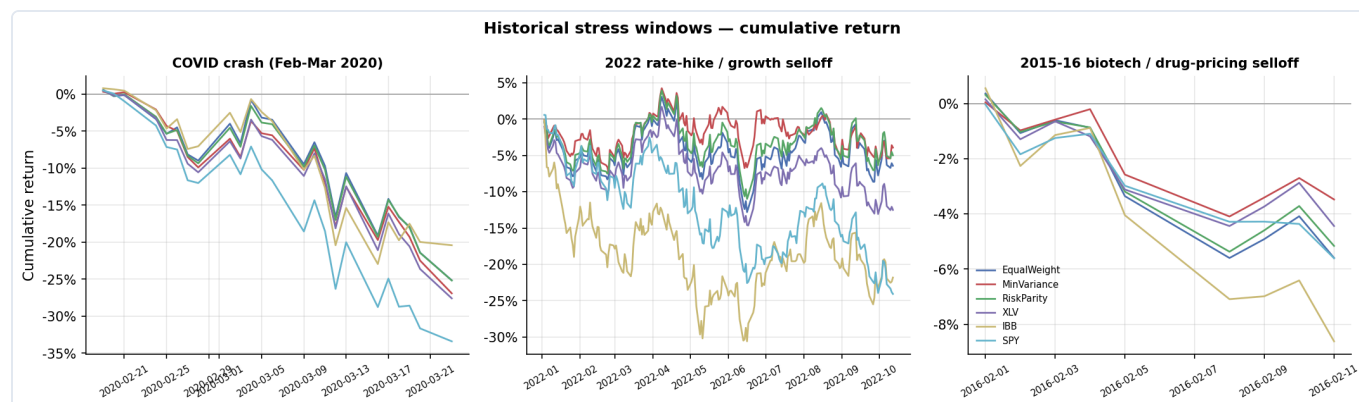


Figure 4. Cumulative return through three stress windows. Diversification within healthcare helps selectively — large in rate-driven and biotech-specific stress, negligible in a broad market crash.

Limitations

A 28-name liquid seed universe (survivorship not fully addressed); in-sample factor betas (only the weights are walk-forward); first-order linear scenarios; and a long-Treasury ETF (TLT) proxy for the rate driver. The fundamental-fragility model is scoped as future work.

Independent research prepared for educational and illustrative purposes. Not investment advice. Data: Yahoo Finance, Kenneth French Data Library. Reproducible end-to-end (Python; tested) — full code and methodology available on request.